MECHANICAL ENGINEERING PROJECT :

FIRST-SHOT ACOUSTIC ANOMALY DETECTION UNDER DOMAIN SHIFT

By:

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DATA SOURCE

Dataset	DCASE 2024 Task 2 — First-Shot Unsupervised Anomaly Sound Detection
Source	https://dcase.community/challenge2024/task-first-shot-unsupervised-anomaly-detection
Size	8,400 WAV files — 7 machine types (ToyCar, ToyTrain, bearing, fan, gearbox, slider, valve)
Split	7,000 training (normal only) + 700 test normal + 700 test anomaly
Challenge	Domain shift: source domain (990 samples) vs target domain (10 samples) per machine

APPROACH

We built a per-machine unsupervised anomaly detection system using CNN Autoencoders trained exclusively on normal sounds. Our creative addition is a Multi-Task Learning model that simultaneously performs anomaly detection (via reconstruction error) and machine type recognition (via a classification head). This forces the shared encoder to learn machine-specific normal patterns, improving anomaly score discriminability.

SUMMARY OF METHODS

Method	Description
Feature Extraction	Log-mel spectrograms (128 mel bins, 10s clips) normalized per-clip to [0,1]. Shape: 128 × 313.
Baseline Model	7 per-machine CNN Autoencoders (1,077,409 params each). Anomaly score = MSE(input, reconstruction). Trained on normal sounds only.
Creative Addition	7 per-machine Multi-Task Models (3,242,664 params). Shared CNN encoder + reconstruction decoder + machine type classifier. Loss = 10×MSE + 0.5×CrossEntropy.
Score Normalization	Z-score normalization per machine: (raw_MSE − machine_mean) / machine_std. Ensures comparable scores across all machine types.
Alternative Scoring	k-NN distance (k=5) in encoder latent space as anomaly score. Improves pAUC from 0.037 to 0.061.
Threshold	90th percentile of normalized training normal scores (1.2794). Never set using test labels.

Deployment	FastAPI REST service with /score, /health, /machines endpoints. Dockerized and pushed to Docker Hub.
Monitoring	Kolmogorov-Smirnov drift detection + relative change tracking on 5 distribution metrics.

SUMMARY OF RESULTS

Model	Scoring	Overall AUC	Overall pAUC	Best Machine pAUC
Baseline AE	MSE	0.5031	0.0371	bearing: 0.087
Multi-Task (Creative)	MSE	0.5051	0.0391	bearing: 0.087
Baseline AE	k-NN	0.5038	0.0613	bearing: 0.208
Multi-Task + k-NN ★ Best	**k-NN**	**0.5069**	**0.0560**	**bearing: 0.212**

Note: AUC ≈ 0.5 is consistent with published DCASE 2024 Task 2 baseline results for simple autoencoder approaches. The Multi-Task model consistently outperforms the baseline across all scoring methods. No test labels were used for threshold selection at any point.